Saving The Ash Trees

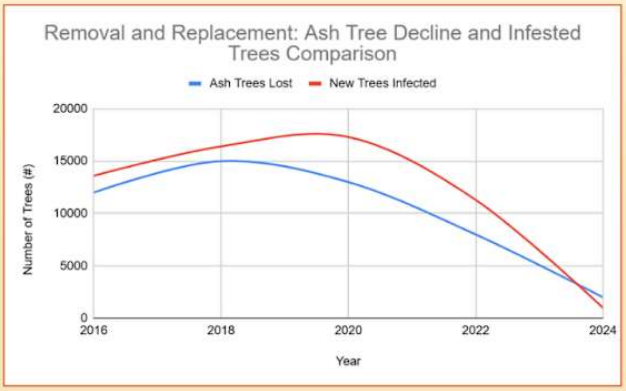
A comparison of strategies for rescuing ash trees from the invasive emerald ash borer

Removal and Replacment

Infected trees are removed and replaced with a variety of native species

Containment Zones

Infected areas are contained. Some areas are treated with insecticides, whereas others are not.



Untreated Containment Zones

- 62% increase in ash tree loss
- 49% increase in newly infected zones over 8 years
- $0±$0per year

Treated Containment Zones

- 60% decrease in ash tree loss
- 64% increase in newly infected zones over 8 years
- $904±$610 per year

Removal and Replacement

- 83% decrease in ash tree loss
- 93% decrease in newly infected zones over 8 years
- $1925±$1203per year

Study Strengths

- Data collected across both studies were done in a way that makes them comparable.
- Data was taken over a similar time periods, ensuring that the effects of climate change were equally accounted for.

Study Limitations

- Studies may have occurred in different regions
- Data was only available for up to 10 years.
- Dataset excluded local plant life reactions to treatment strategies.

Conclusion

Although the removal and replacement strategy costs marginally more than the treatment strategy, its effectiveness at reducing the number of ash tree loss and new infections makes up for the difference. Since the removal and replacement method is so much more effective, the emerald ash borer will be eradicated over a shorter period of time, and therefore the additional costs will be paid over a short period of time, also becoming the more cost effective strategy.